Suricata IDS Practical Lab

Beginner-Friendly • Copy-Paste Ready • Authorized Cybersecurity Lab

Overview & Objectives

Install and configure Suricata; identify the monitored interface; create a local TCP/4444 SYN rule; generate Netcat traffic; trigger and analyze alerts; use grep and jq; troubleshoot common failures.

Topology

CLIENT: → lab network → SURICATA SENSOR: , monitored interface: . Use only authorized systems.

Prerequisites

Kali/Debian/Ubuntu sensor, authorized client, sudo access, network connectivity, apt access, basic terminal skills.

1. Identify the Network Interface

ip -br addr shows interfaces, state, and addresses. ip route shows which interface reaches the lab subnet. Ping provides a basic reachability check.

```
ip -br addr
```

```
ip route
```

```
ping -c 3 <CLIENT_IP>
```

2. Install Suricata

The package manager installs the IDS and analysis/test utilities. Version/help commands verify that the programs are available.

```
sudo apt update
```

```
sudo apt install -y suricata jq netcat-openbsd
```

```
suricata -V
```

```
jq --version
```

```
nc -h
```

3. Validate Configuration

A successful test means Suricata can parse its configuration and rules. Exact success text varies by version.

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

4. Configure HOME_NET and Capture

Back up before editing. Set HOME_NET to the actual lab subnet and configure the correct capture interface according to the installed Suricata version. Validate again with suricata -T.

```
sudo cp /etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.bak
```

```
sudo nano /etc/suricata/suricata.yaml
```

5. Create the Custom Rule

The rule alerts on TCP SYN packets headed to TCP/4444 in HOME_NET. SID 1000001 identifies this lab signature.

```
sudo nano /etc/suricata/rules/local.rules
```

```
alert tcp any any -> $HOME_NET 4444 (msg:"LAB Netcat TCP 4444 Test"; flags:S; sid:1000001; rev:1;)
```

6. Generate Netcat Traffic

The listener provides a test service. The client-side connection test generates a TCP connection attempt. A refused connection can still expose a SYN packet for IDS detection.

```
nc -lvnp 4444
```

```
nc -vz <SURICATA_IP> 4444
```

7. Start Suricata

Run in the foreground during the lab so startup/configuration problems are visible. Keep it running while generating traffic.

```
sudo suricata -c /etc/suricata/suricata.yaml -i <INTERFACE>
```

8. Watch and Search Alerts

fast.log is a human-readable alert log. grep searches the signature message or SID.

```
sudo tail -f /var/log/suricata/fast.log
```

```
sudo grep -n "LAB Netcat TCP 4444 Test" /var/log/suricata/fast.log
```

```
sudo grep -n '1000001' /var/log/suricata/fast.log
```

9. Analyze eve.json with jq

eve.json stores structured JSON events. jq can filter alert events and extract fields for investigation.

```
sudo jq 'select(.event_type=="alert" and .alert.signature_id==1000001)' /var/log/suricata/eve.json
```

```
sudo jq -r 'select(.event_type=="alert" and .alert.signature_id==1000001) | [.timestamp,.src_ip,.src_port,.dest_ip,.dest_p
```

10. Verify Packet Visibility

tcpdump proves whether packets reach the monitored interface. pgrep confirms the process and ls confirms log files exist.

```
sudo tcpdump -ni <INTERFACE> tcp port 4444
```

```
pgrep -a suricata
```

```
sudo ls -lah /var/log/suricata/
```

Rule

```
alert tcp any any -> $HOME_NET 4444 (msg:"LAB Netcat TCP 4444 Test"; flags:S; sid:1000001; rev:1;)
```

Troubleshooting

No route to host

Check topology, addressing and routes.

```
ip addr
```

```
ip route
```

```
ping -c 3 <CLIENT_IP>
```

Connection refused

Start the listener or confirm the intended rejection; a SYN can still trigger the IDS rule.

```
nc -lvnp 4444
```

```
nc -vz <DESTINATION_IP> 4444
```

Wrong interface

Select the interface that carries the lab traffic.

```
ip -br addr
```

```
ip route
```

```
sudo tcpdump -ni <INTERFACE> tcp port 4444
```

Missing alerts

Validate configuration/rules, prove packet visibility, and verify the process.

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

```
sudo tcpdump -ni <INTERFACE> tcp port 4444
```

```
pgrep -a suricata
```

Verification Checklist

■ Interface identified

■ Suricata installed

■ Configuration validates

■ HOME_NET correct

■ SID 1000001 loaded

■ Netcat traffic generated

■ tcpdump sees packets

■ fast.log alert

■ eve.json alert

■ grep/jq analysis

■ Evidence sanitized

Student Questions

1. What interface was monitored and why?
2. What is the purpose of HOME_NET?
3. What does flags:S match?
4. What does SID 1000001 identify?
5. What source/destination IP and port appeared?
6. Why can an alert appear when Netcat says Connection refused?
7. What is the difference between fast.log and eve.json?
8. Why is jq useful for eve.json?
9. What evidence proves the packet reached the sensor?
10. How would you change the rule to monitor another port?

Practical Challenge

Create a TCP/5555 SYN rule with SID 1000002, generate authorized Netcat traffic, validate fast.log/eve.json, and extract key fields with jq.